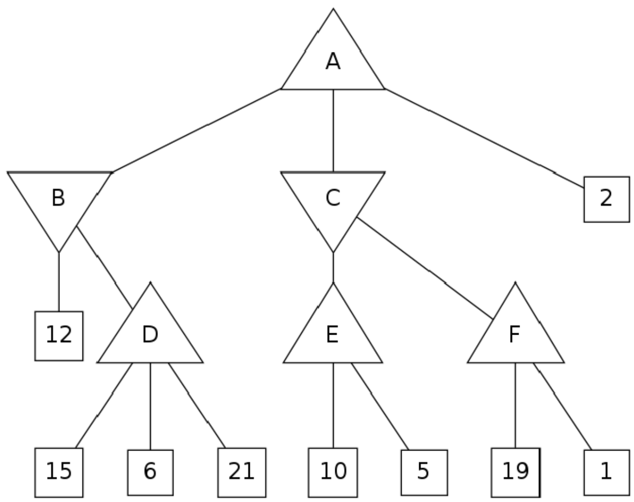
CS 4470: Artificial Intelligence

HW 2

Points: Please see the points for each problem.
Instruction: Please submit on canvas

1. Consider the minimax tree shown below for parts a and b.

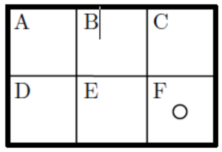


a. What value will root node A have? (4pts)

b. Cross off the nodes that are pruned by $\alpha\beta$ pruning. Assume the standard left-to-right traversal of the tree. If a non-terminal state (A, B, C, D, E, or F) is pruned, cross off the entire subtree. (4pts)

2. The following a-c refer to the environment below.
Pacman is using MDPs to maximize his expected utility. In each environment:
   - Pacman has the standard actions {North, East, South, West} unless blocked by an outer wall. F is the goal state, so the only available action is TERMINATE.
   - There is a reward of 1 point when eating the dot. For example, in the grid below, R([C, South, F]) = 1)
   - The game ends when the dot is eaten

Consider the following grid where there is a single food pellet in the bottom right corner F. The discount factor is 0.5. Living reward is 0. The states are simply the grid locations.



a. What is the optimal policy for each state A-F? (4pts)

b. What is the optimal value $V^*(A)$ for state A? (4pts)
V*(A) = ?

c. Using value iteration with the value of all states equal to zero at time $k = 0$, at which time step k will $V_k(A) = V^*(A)$? (4pts)
k=?